

Game Theory

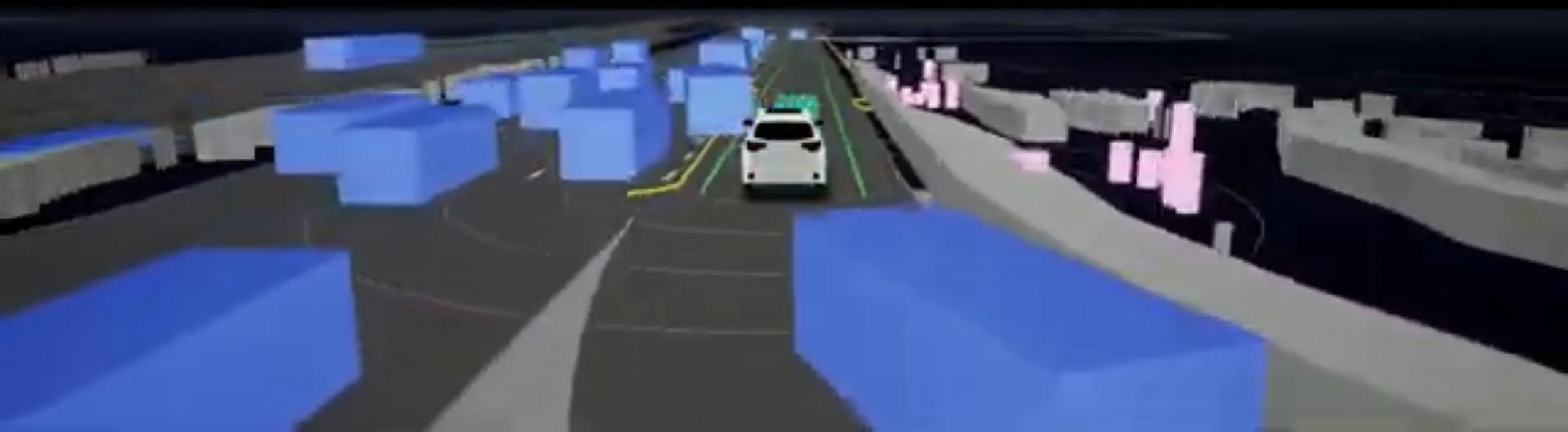
Dylan Losey



2018/08/09 08:56:39

A 3X SPEED

ZOOX



This Lecture



- How do we write human-robot interaction as a two-player game?
- Using game theory to influence humans

Prisoners' dilemma

		prisoner B			
		confess		remain silent	
prisoner A	confess	 5 years	 5 years	 0 year	 20 years
	remain silent	 20 years	 0 year	 1 year	 1 year



How can we write HRI as
a **two-player game**?

Initial state s^0



Multi-Agent Games

Consider a human-robot system with **dynamics**:

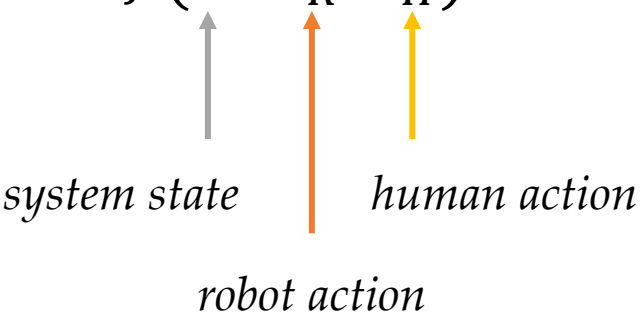
$$s^{t+1} = f(s^t, a_R^t, a_H^t)$$


Diagram illustrating the dynamics equation $s^{t+1} = f(s^t, a_R^t, a_H^t)$. The inputs are labeled below the equation:

- s^t is labeled *system state* (indicated by a grey arrow).
- a_R^t is labeled *robot action* (indicated by an orange arrow).
- a_H^t is labeled *human action* (indicated by a yellow arrow).

Multi-Agent Games

Consider a human-robot system with **dynamics**:

$$s^{t+1} = f(s^t, a_R^t, a_H^t)$$

Let's assume the dynamics are **deterministic**. Over T timesteps:

- $\mathbf{a}_R = (a_R^0, a_R^1, \dots, a_R^T)$ is the sequence of robot actions
- $\mathbf{a}_H = (a_H^0, a_H^1, \dots, a_H^T)$ is the sequence of human actions

Sequence of robot actions



Sequence of human actions



Multi-Agent Games

The human and robot have **separate** reward functions

$$R_H(s^0, \mathbf{a}_R, \mathbf{a}_H) = \sum_{t=0}^T r_H(s^t)$$

$$R_R(s^0, \mathbf{a}_R, \mathbf{a}_H) = \sum_{t=0}^T r_R(s^t)$$

Multi-Agent Games

The human and robot have **separate** reward functions

1. **Collaboration**. Both agents get same reward at every timestep.

$$R_H(s^0, \mathbf{a}_R, \mathbf{a}_H) = R_R(s^0, \mathbf{a}_R, \mathbf{a}_H)$$

2. **Competition**. Agents get different rewards.

$$R_H(s^0, \mathbf{a}_R, \mathbf{a}_H) \neq R_R(s^0, \mathbf{a}_R, \mathbf{a}_H)$$

Often we have a mix of
collaboration and **competition**.



Slow down
human
and avoid
collisions



Avoid collisions
and go fast

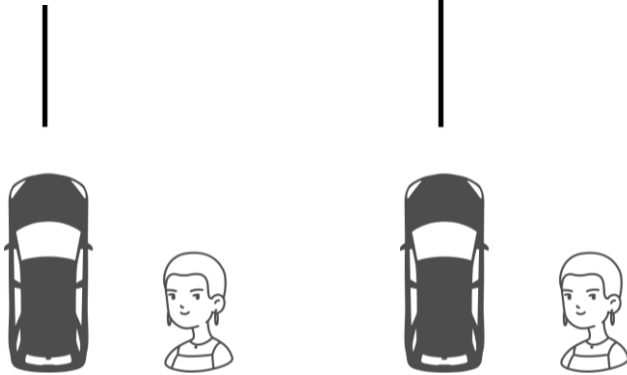
A green table football table with white lines. Several player figures are on the rods. The figures are in red and white, yellow, and blue. The text "Robots should reason over how humans will react." is overlaid on the image.

Robots should reason over
how humans will **react**.

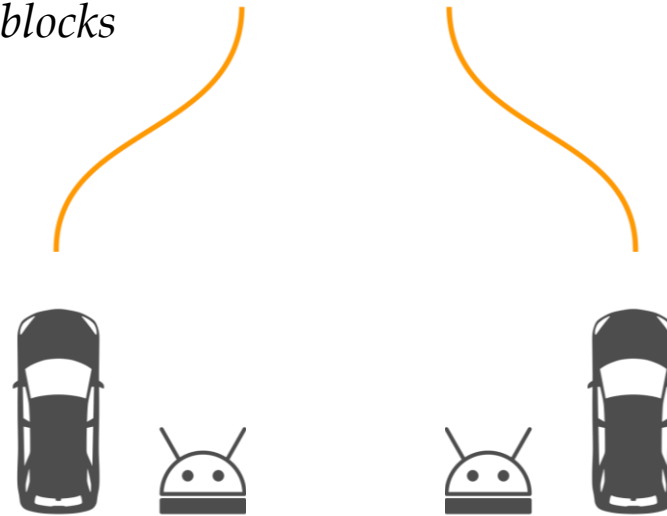
Nash Equilibrium

For now let's simplify the problem so that each agent takes a single action.

H goes slow or fast



R yields or blocks



Nash Equilibrium

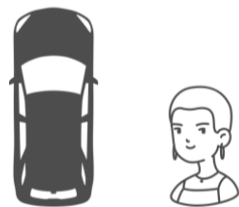
Nash Equilibrium.

Pair of human and robot policies where, if either agent changes their policy, they will receive less expected reward.

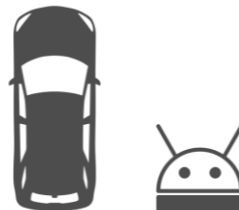
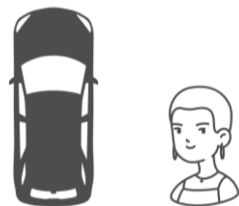
Notes.

- For a given problem there can be more than one Nash Equilibrium
- When we are uncertain about the other agent's policy, we play the policy that works best in expectation

Human: Fast



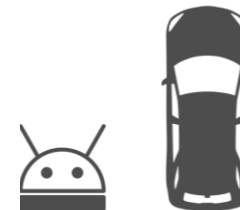
Human: Slow



Robot: Get out of way

$$R_H = +10$$

$$R_R = +0$$



Robot: Block human

$$R_H = -100$$

$$R_R = -100$$

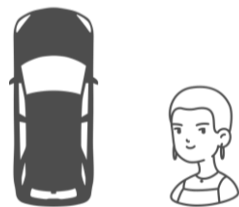
$$R_H = +5$$

$$R_R = +5$$

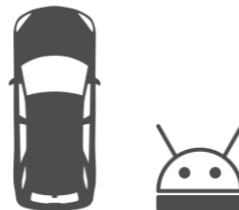
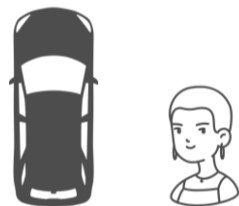
$$R_H = +1$$

$$R_R = +7$$

Human: Fast



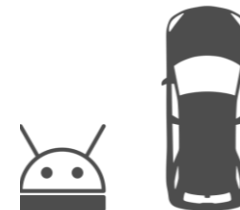
Human: Slow



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Robot: Block human

$$R_H = -100$$

$$R_R = -100$$

$$R_H = +0$$

$$R_R = +0$$

$$R_H = +1$$

$$R_R = +10$$



How can robots
plan while
accounting for
human
responses?



Robot optimizes for
 $R_R(s^0, \mathbf{a}_R, \mathbf{a}_H)$

Sequence of robot actions



Human optimizes for
 $R_H(s^0, \mathbf{a}_R, \mathbf{a}_H)$

Sequence of human actions



Stackelberg Games

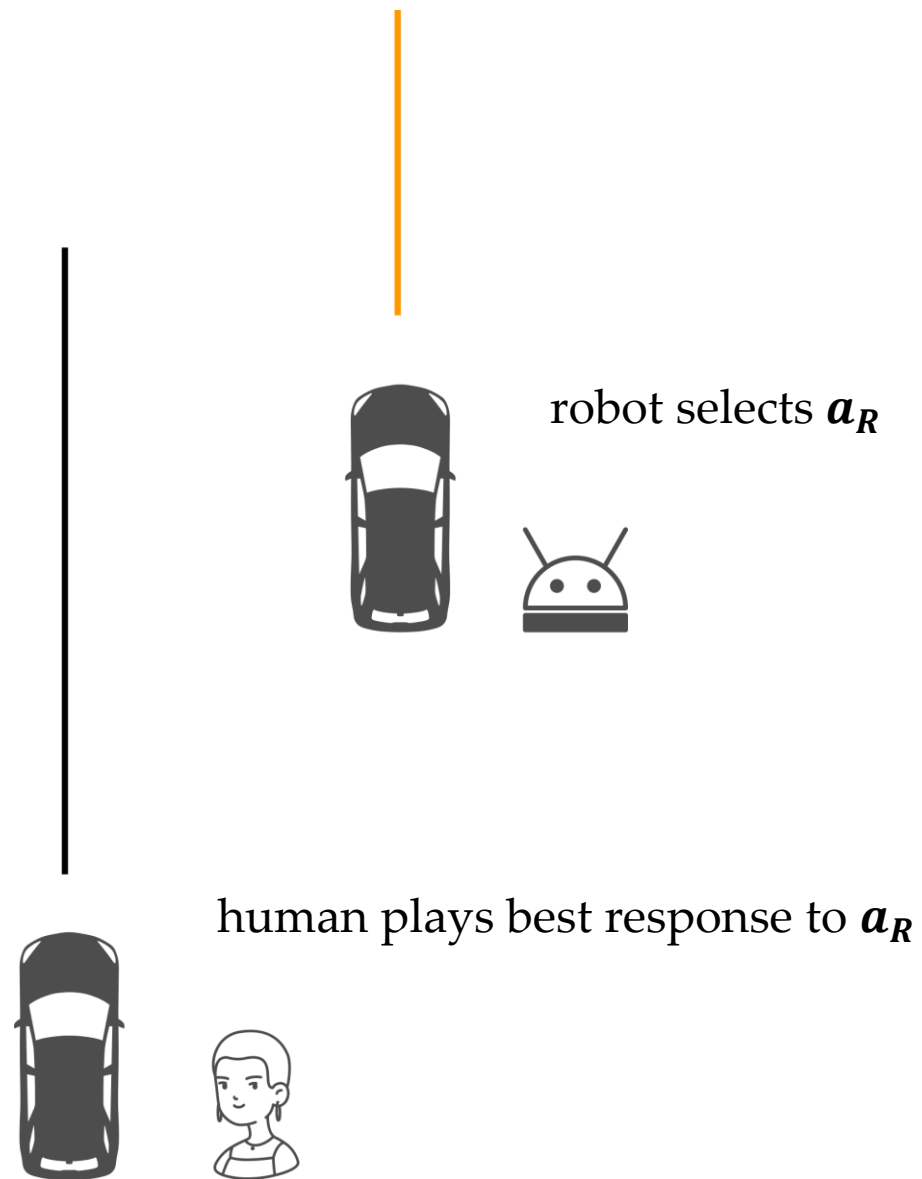
In a **Stackelberg game** one agent gets to play first.

The other agent observes their actions, and then chooses a best response.



robot selects $\mathbf{a}_R$





Stackelberg Games

In **Stackelberg game** robot plays first and human chooses best response.

$$\mathbf{a}_H^* = \operatorname{argmax}_{\mathbf{a}_H} R_H(s^0, \mathbf{a}_R, \mathbf{a}_H)$$

Given robot selection $\mathbf{a}_R$
human finds optimal actions $\mathbf{a}_H^*$
Note that $\mathbf{a}_H^*$ depends on the robot's choice $\mathbf{a}_R$

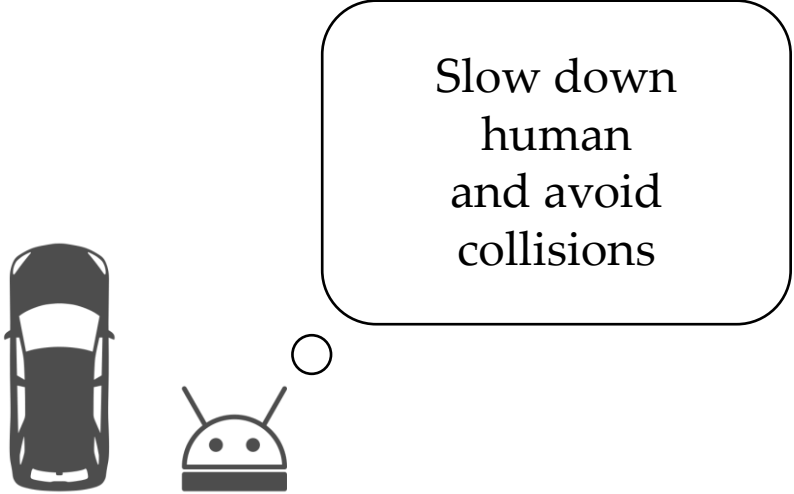
Stackelberg Games

In **Stackelberg game** robot plays first and human chooses best response.

$$\mathbf{a}_H^* = \operatorname{argmax}_{\mathbf{a}_H} R_H(s^0, \mathbf{a}_R, \mathbf{a}_H)$$

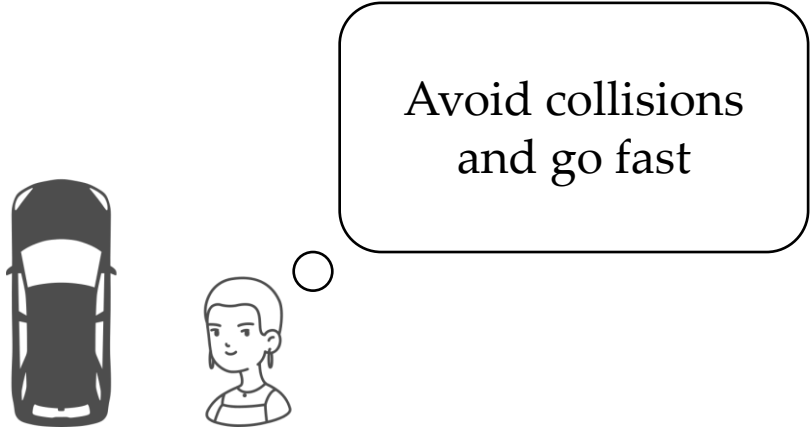
$$\mathbf{a}_R^* = \operatorname{argmax}_{\mathbf{a}_R} R_R(s^0, \mathbf{a}_R, \mathbf{a}_H^*(\mathbf{a}_R))$$

Robot chooses $\mathbf{a}_R$ that leads to best human response (i.e., response where the robot can get the most reward).



Slow down
human
and avoid
collisions

A top-down view of a black car is positioned to the left of an Android robot. A small circle connects the robot to a large speech bubble containing the text 'Slow down human and avoid collisions'.



Avoid collisions
and go fast

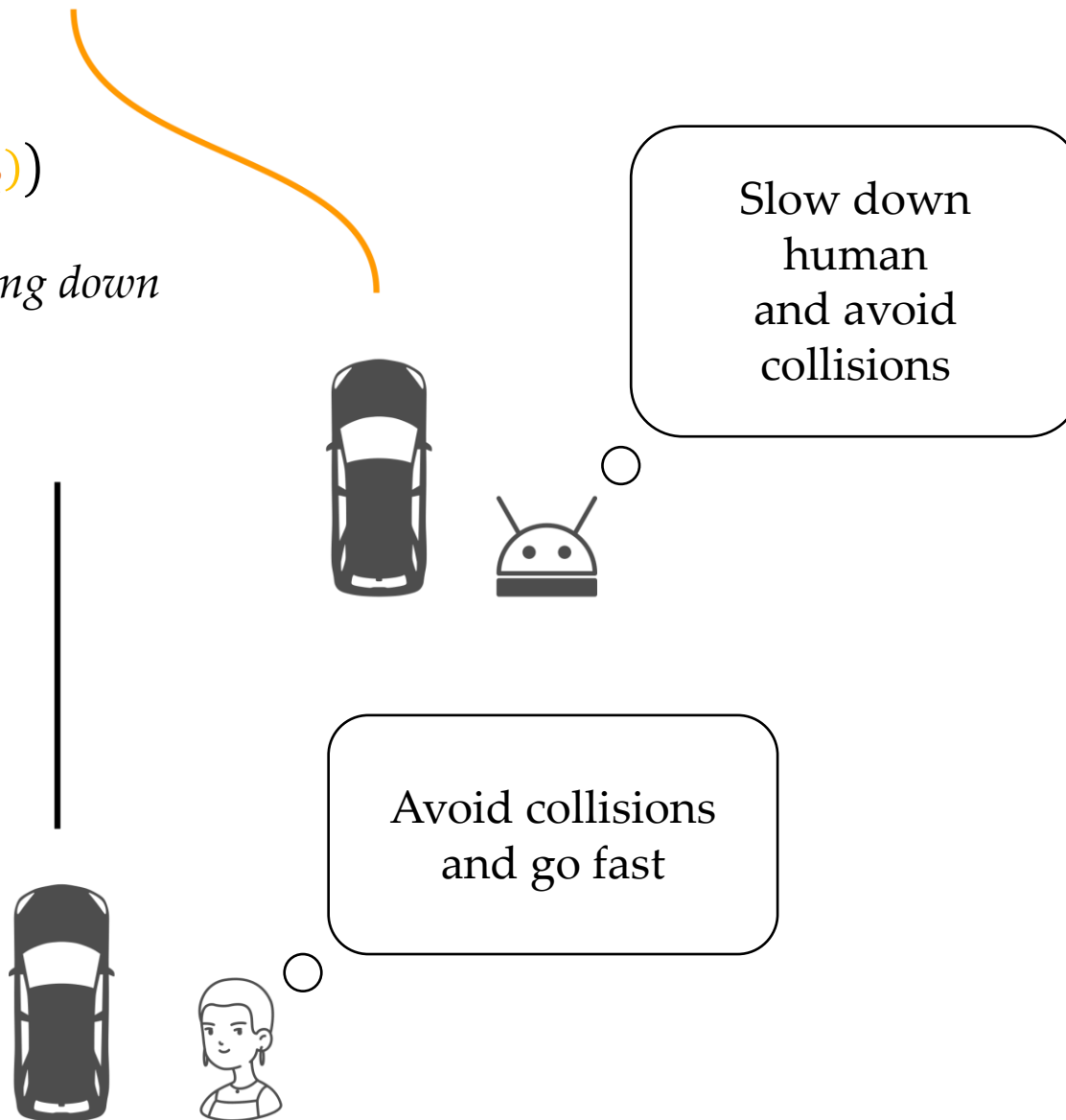
A top-down view of a black car is positioned to the left of a person's head and shoulders. A small circle connects the person to a large speech bubble containing the text 'Avoid collisions and go fast'.

$$\mathbf{a}_R^* = \operatorname{argmax}_{\mathbf{a}_R} R_R(s^0, \mathbf{a}_R, \mathbf{a}_H^*(\mathbf{a}_R))$$

Block human, their best response to slowing down

$$\mathbf{a}_H^* = \operatorname{argmax}_{\mathbf{a}_H} R_H(s^0, \mathbf{a}_R, \mathbf{a}_H)$$

Slow is best response to blocking





Robots that plan for
human response can
influence humans

Influence

Consider a human-robot setting where the robot gets trajectory reward R

Influence **naturally emerges** as part of the robot's solution when:

1. The robot has a model of human
2. The human changes behavior in response to robot actions

Influence

Consider a human-robot setting where the robot gets trajectory reward R

Influence **naturally emerges** as part of the robot's solution when:

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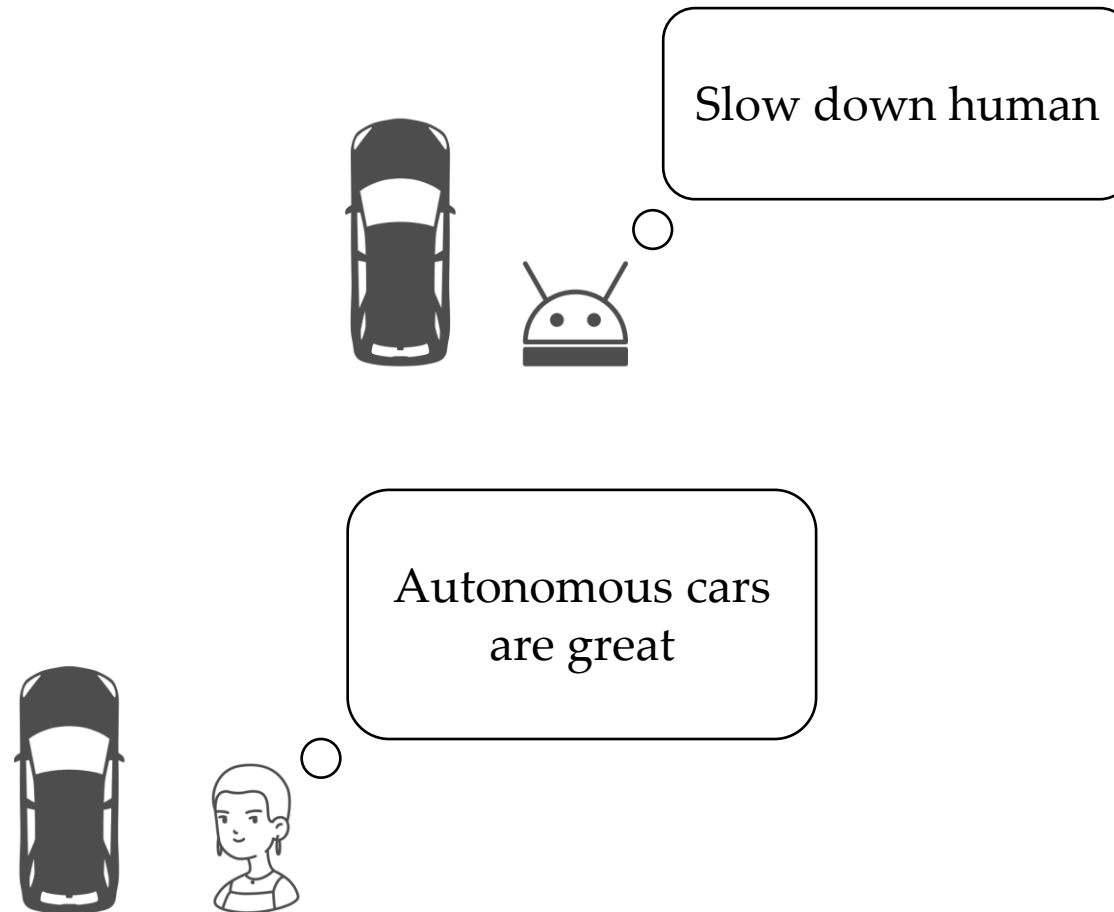
$$\mathbf{a}_H = M(s^0, \mathbf{a}_R)$$

$$\mathbf{a}_R^* = \operatorname{argmax}_{\mathbf{a}_R} R_R(s^0, \mathbf{a}_R, M(s^0, \mathbf{a}_R))$$

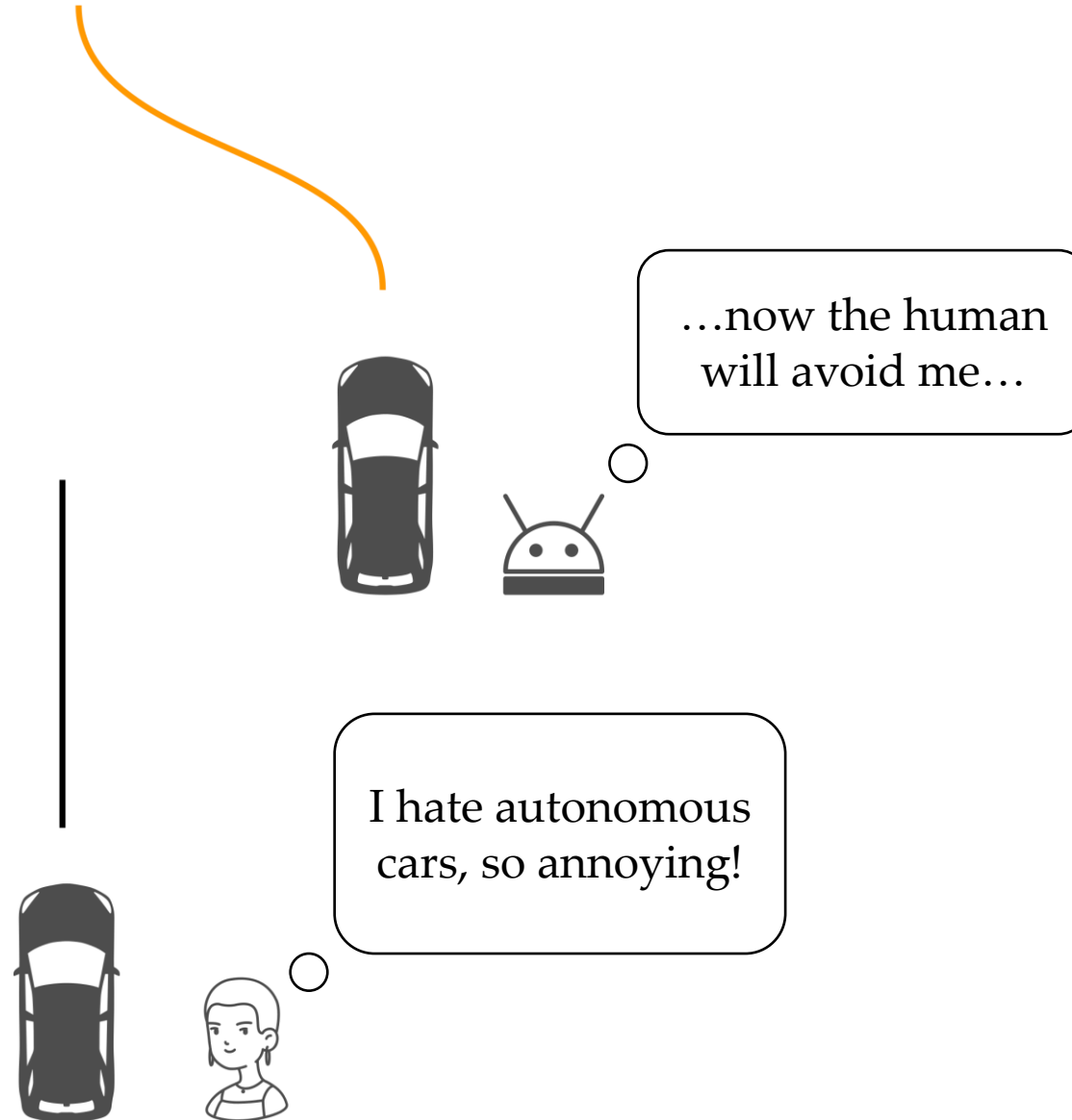
A stylized, semi-transparent robotic arm and tripod stand in the background. The arm is grey and has a circular sensor or camera lens on its upper section. It is positioned vertically, with the tripod stand visible at the bottom. The entire scene is set against a solid brown background.

How does influence
affect **trust**?

Influence



Influence



Towards Robots that Influence Humans over Long-Term Interaction

Shahabedin Sagheb, Ye-Ji Mun, Neema Ahmadian, Benjamin A. Christie
Andrea Bajcsy, Katherine Driggs-Campbell, and Dylan P. Losey



This Lecture



- How do we write human-robot interaction as a two-player game?
- Using game theory to influence humans

Next Lecture



- Machine teaching + topics you requested